

TDC Module (CAEN V1190B)

TDC Module (Time to Digital Converter V1190B)

The Model V1190B is a 1-unit wide VME 6U module that houses 64 independent Multi-Hit/Multi-Event Time to Digital Conversion channels.

The unit houses 4 high Performance TDC chips, developed by CERN/ECP-MIC Division. Resolution can be set at 100 ps (19 bit dynamics, 52 us FSR), 200ps(19 bit dynamics, 104 us FSR) or 800ps(17 bit dynamics, 104 us FSR). The board houses a 32 kwords deep Output Buffer, that can be readout via VME(as single data, Block Transfer and Chained Block Transfer) in a completely independent way from the acquisition itself.

Note: trigger resolution for this unit is 25ns. If acquisition mode 'trigger matching' is used then you need to have delayed trigger signal also presented on one of data input of TDC.

For our goals we have used next settings for TDC operation:

- resolution is 100 ps
- detected edge is leading
- number of hits is 16
- acquisition mode: trigger matching
- enable trigger time subtraction
- no DMA mode, only FIFO readout
- in CODA *.crl file only low 16 bits of data are used
- base address is 0x10200000 (on board rotary switch: 0x08200000)

Configuring of TDC for operation.

To configuring logic of TDC operations routines from 'tdc890.o' library(located in "adaqel2:/home/moller/dev/v1190/") are used.

This can be done in the coda *.crl file (located in "adaqel2:/home/moller/coda/crl/") or using VxWorks terminal.

The file 'tdc890.o' should be downloaded to VME crate using command from VxWorks terminal:

```
ld < path_to_file/tdc890.o
```

The detailed description of the functions one can find in source file 'tdc890.c' located in "adaqel2:/home/moller/dev/v1190/".

Compilation of *.crl file for using with CODA.

To compile '*.crl' code one can type (under moller account on adaqel2) :

```
tcsh
```

```
cd coda2/crl
```

```
makelist test_TDC.crl ppc
```
